

CRANWELL, United Kingdom

WMO: 033790

Lat: 53.0311N

Lon: 0.5036W

Elev: 66

StdP: 100.53

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.4	-2.1	-6.0	2.3	-0.7	-4.7	2.6	-0.3	16.8	9.0	15.1	9.3	3.5	250	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.3	18.3	24.2	17.5	22.5	16.6	19.2	24.1	18.2	22.7	17.4	21.3	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	21.1	16.6	11.9	20.0	15.7	11.3	19.0	54.9	24.1	51.7	22.7	49.1	21.4	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	12.1	10.8	DB	-6.4	30.3	1.9	1.9	-7.7	31.6	-8.8	32.7	-9.9	33.7	-11.2	35.1	
			WB	-6.8	22.0	1.8	2.0	-8.1	23.5	-9.2	24.6	-10.2	25.7	-11.5	27.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.1	4.5	6.3	8.6	11.5	14.5	16.9	16.8	14.2	10.8	6.9	4.5
	DBStd	5.42	3.08	3.23	3.01	2.94	2.89	2.61	2.56	2.43	2.33	2.81	3.00	3.49
	HDD10.0	829	182	155	120	60	17	1	0	0	1	23	99	171
	HDD18.3	3077	441	387	374	291	211	119	60	62	127	234	343	428
	CDD10.0	829	1	2	4	19	64	134	214	209	125	48	7	2
	CDD18.3	36	0	0	0	0	0	4	16	13	2	1	0	0
	CDH23.3	271	0	0	0	1	4	38	124	89	13	2	0	0
	CDH26.7	46	0	0	0	0	0	6	21	18	1	0	0	0
Wind	WSAvg	5.3	6.1	6.1	5.7	5.1	4.9	4.6	4.5	4.6	4.9	5.4	5.6	5.9
Precipitation	PrecAvg	583	51	36	44	47	46	53	46	60	51	49	54	52
	PrecMax	726	112	132	103	114	120	144	147	135	137	118	106	146
	PrecMin	398	11	5	6	7	5	2	9	3	7	5	23	13
	PrecStd	93	25	24	23	26	27	36	30	33	35	31	22	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.9	16.2	21.0	23.4	27.5	29.1	29.0	25.2	20.3	15.5	13.6
		MCWB	11.1	11.0	11.1	13.9	16.6	19.3	19.3	18.7	17.2	16.1	13.7	11.7
	2%	DB	11.3	12.0	14.2	17.7	21.1	23.9	26.5	25.8	22.3	17.8	13.9	12.4
		MCWB	9.6	9.8	10.2	12.2	15.5	17.0	18.4	18.3	16.8	14.6	12.3	11.0
	5%	DB	10.1	10.7	12.7	15.6	19.0	21.6	24.5	23.4	20.2	16.4	12.7	11.1
		MCWB	8.6	8.7	9.6	11.1	14.0	16.0	17.6	17.2	15.5	13.9	11.1	9.8
	10%	DB	8.9	9.5	11.3	13.9	17.1	19.8	22.5	21.7	18.6	15.2	11.4	9.8
		MCWB	7.6	7.7	8.6	10.1	12.8	15.0	16.6	16.3	14.5	12.9	10.1	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.5	12.3	14.7	17.7	20.1	21.0	20.4	18.9	16.7	14.0	12.0
		MCDB	12.6	13.3	14.8	19.5	22.1	25.9	27.6	25.4	22.8	19.4	15.2	13.0
	2%	WB	9.9	10.0	11.0	13.1	15.8	17.9	19.2	19.0	17.3	15.4	12.6	11.1
		MCDB	11.1	11.7	13.4	16.7	20.2	22.1	24.4	23.7	20.9	17.2	13.7	12.3
	5%	WB	8.8	9.0	9.9	11.7	14.6	16.6	18.2	18.1	16.3	14.4	11.4	10.0
		MCDB	9.9	10.5	12.2	14.9	18.2	20.4	22.9	22.3	19.1	15.9	12.4	11.0
	10%	WB	7.7	7.9	8.9	10.5	13.4	15.7	17.3	17.2	15.3	13.3	10.3	8.7
		MCDB	8.7	9.3	11.0	13.4	16.4	19.0	21.4	20.7	17.7	14.8	11.2	9.7
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	7.2	8.4	8.7	8.7	9.3	8.9	8.1	6.8	5.6	4.9
		MCDBR	6.4	6.9	9.2	10.9	11.8	11.9	12.7	11.6	10.4	7.7	6.2	6.0
		MCWBR	5.8	5.5	6.0	6.4	6.9	6.3	5.8	5.2	5.8	5.3	5.3	5.5
	5% WB	MCDBR	6.4	6.6	8.3	10.2	10.8	10.8	11.3	10.3	9.4	7.5	6.1	5.9
		MCWBR	5.9	5.6	6.0	6.4	6.8	6.4	5.8	5.2	6.0	5.7	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.323	0.339	0.371	0.397	0.399	0.403	0.404	0.392	0.370	0.351	0.325	0.311	
	taud	2.359	2.388	2.294	2.231	2.267	2.277	2.284	2.344	2.406	2.442	2.409	2.335	
	Ebn at Noon	653	765	808	832	850	848	840	830	804	740	648	590	
	Edh at Noon	59	77	103	124	126	126	123	110	92	73	56	51	
All-Sky Solar Radiation	RadAvg	0.71	1.39	2.46	3.80	4.73	4.97	4.81	3.97	2.95	1.70	0.87	0.57	
	RadStd	0.07	0.16	0.30	0.44	0.31	0.48	0.51	0.27	0.27	0.15	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+42	N/A
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page